FINAL EXAM

Fall 2021

- This is a closed-book exam. Only a 8.5"x11" formula sheet is allowed.
- No electronic devices aside from a basic 5 function calculator that is provided.
- A camera will be set in the classroom to monitor exam situations (Exam recordings will be deleted once final grades are posted unless required for an ongoing Honor Committee case).
- If you have clarifying questions, Help Desks can be found at each of the "four wings (#2200, #2400, #4200, #4400)" of the building. Faculty will also circulate in each classroom throughout the exam.
- There are <u>180 points</u> in the exam. You also have <u>180 minutes</u>. DO NOT speak to others before handing in your exam booklets.
- Complete all SHORT ANSWER questions in one Blue Book, and all LONG PROBLEMS in a 2nd Blue Book. Write your name and "SHORT" or "LONG" on the cover of your Blue Books.
- Please be as complete as possible. Write down the reasonings for your answer (unless instructed otherwise).
- You may not communicate with other students during the exam. You will also keep the exam contents confidential.
- Structure:
    - Short questions: 54 points
    - Econ problems: 81 points
    - Game theory problems: 45 points

Section I: Short Questions [54 points]

Please be concise. None of the answers require more than a brief paragraph. All questions are worth 9 points each.

1. In the US, the president has, at many times over the past few decades, imposed import tariffs on different goods. List two reasons why economists have generally had a negative view on the effects of import tariffs. (There exist views in support, but we are asking for reasons why economists would generally oppose these tariffs, which is the case).

Answer: Deadweight loss, domestic consumers are harmed, and this can increase input costs for domestic manufacturers.

2. An airline recently increased fares on the BDL-ORD route by 10% and noticed that bookings fell by approximately 8%. What is the (precise) elasticity of demand on this route at the original price and would you describe it as elastic or inelastic?

Answer: elasticity = -0.08/0.1 = -0.8, inelastic

3. Amid increasing competition from Uber and Lyft, Unique Taxi/Cab---a New Haven based taxi company---is considering a new business model to attract customers. They plan to introduce a membership program that gives consumers who pay a one-time membership fee unlimited lifetime rides. If you are a consultant of this company, would you support this plan or would you advise against it? Why?

Answer: Against, because the marginal cost of a ride is significantly greater than zero.

4. A local brewery uses both machines (m) and labor (l) to make their product. Their production function is
$$f(m, l) = 2m + l + 0.5ml.$$
Suppose that there is a new generation of machines that are twice as efficient, which leads to a new production function
$$g(m, l) = 4m + l + ml.$$
If the company wants to maintain the same output in a day, how would the number of machines and number of employees change after this innovation? (Assume no changes in the costs of either labor or machines).

Answer: MPIs' have to be the same, more machines, fewer employees.

5. A company that sells diet supplements on the Internet recently conducted a price experiment on one of their best-sellers, a multi-vitamin that promotes healthy digestion. They currently sell a bottle of capsules for $5 that they manufacture at a cost of $2.50. They tried increasing the price to $6 in some parts of the country based on a shopper's IP address and found that demand fell about 10% when they did so. Is $5 the optimal price for this product?

Answer: Demand is inelastic and you shouldn't price in the inelastic region; IEPR doesn't hold.

6. A year ago, a local clothing store commissioned a consultant to help them design an optimal wage and bonus structure for their employees. The outcome was that they paid their employees $15 per hour plus a commission of $5 each time a customer made a purchase. However, recent upward pressure on wages means that the next best option for most of their staff now pays a flat hourly wage that is $2 more per hour than when the consultant had devised this salary system. What – if any – changes would you recommend to the current wage and bonus structure?

Answer: the base wage needs to increase by $2. So, $17 per hour and $5 commission.

Section II: PROBLEMS (81 points)

Problem 1 (14 points)

Demand for surgical masks soared during the pandemic. Disposable, single-use surgical masks are easy to manufacture and are supplied by many small firms. Siftex, a typical firm in this industry that manufacturers masks in Connecticut, has the following total cost function (in thousands of dollars) to manufacture Q (in thousands) boxes of masks in a week:

$$TC(Q) = 1 + Q + 0.1Q^2$$

a) (5 points) Prior to the pandemic, a box of surgical masks would go for \$3. At that price, how many boxes per week would they produce, and what were weekly profits?

Quantity (thousands):

Weekly Profits (\$thousands):

$$MC(Q) = 1 + 0.2Q$$
$$3 = 1 + 0.2Q$$
$$Q = 10$$
$$\pi = 10 * 3 - 1 - 10 - 0.1(100) = 9$$

b) (4 points) When the pandemic hit, demand for masks increased. All sorts of business were calling Siftex and inquiring about masks. The market price for a box of masks increased to the point that Siftex found it optimal to produce 38 thousand boxes per week. What must the market price have been?

Market price for a box of masks:

$$1 + 0.2(38) = 8.6$$

c) (5 points) As the weeks dragged on, other local firms converted their operations to making surgical masks as well to meet the soaring demand. What would we expect the long-run equilibrium price to be for a box of surgical masks once we allow for entry and exit?

Long-run equilibrium price:

$$MIN(ATC)$$
$$\frac{1}{Q} + 1 + 0.1Q = 1 + 0.2Q$$
$$10 = Q^2$$
$$Q = 3.16$$
$$MC(10) = 1 + 0.2 * 3.16 = \$1.63$$

Problem 2 (18 points)

Marriott encourages its loyal customers to always search and book on their website, where they could potentially use tracking cookies and consumers' browsing history to price discriminate against them. Suppose that there are two groups of consumers interested in a hotel in Paris: Business and Leisure, denoted by B and L respectively. The demands for these two groups are given by:

$$Q_B = 1000 - P$$
$$Q_L = 1500 - 2P$$

a) (7 points) Suppose that the marginal cost for Marriott in Paris is \$200 per room. What are the profit-maximizing prices for each group of consumers? What are the quantities sold to each group under these prices?

Answer: Inverse demands are

$$P_B = 1000 - Q_B$$
$$P_L = 750 - 0.5Q_L$$

Marginal revenues are

$$MR_B = 1000 - 2Q_B$$
$$MR_L = 750 - Q_L$$

Therefore, optimal prices and quantities are

$$P_B = 600, \qquad Q_B = 400$$
$$P_L = 475, \qquad Q_L = 550$$

In 2018, the European Union implemented the General Data Protection Regulation (GDPR) and prohibited websites to collect cookies from consumers without their consents. As a result, Marriott cannot distinguish users anymore and thus cannot price discriminate them.

b) (7 points) Assuming that marginal costs remain unchanged, what is Marriott's profit-maximizing price and quantity after the implementation of the GDPR? How have quantities changed? (Hint: Can use fractions as answers, no need to convert to decimals)

Answer: Horizontally adding up the demands:

$$Q = \begin{cases} 2500 - 3P, & P < 750 \\ 1000 - P, & P \geq 750 \end{cases}$$

The inverse demand is then given by:

$$P = \begin{cases} 1000 - Q, & Q < 250 \\ 833.33 - 0.33Q, & Q \geq 250 \end{cases}$$

Marginal revenue is:

$$MR(Q) = \begin{cases} 1000 - 2Q, & Q < 250 \\ 833.33 - 0.67Q, & Q \geq 250 \end{cases}$$

With price discrimination, $P_B = 600 < 750$ from part a), so not optimal to sell to business alone.

Now set $MR(Q) = 833.33 - 0.67Q = MC = 200$, optimal quantity is Q*=950 and optimal price is P*=516.67. Quantities sold remain unchanged.

c) (4 points) Has the introduction of GDPR caused consumer surplus to go up or down among Business travelers? Among Leisure travelers? (No need to give exact numbers, just up or down)

Answer: Business up, Leisure down.

Problem 3 (17 points)

New Haven has succumbed to the trend and introduced a "sweetened beverage" tax, just like cities such as Philadelphia and Berkeley. The original demand for sweetened beverages in the Elm City is given by:

$$P_D = 6.2 - 0.2Q$$

Where P is the price of a beverage and Q is the quantity available in the city (in thousands).

Supply is fairly elastic given the ease of bottling anywhere in the northeast. The industry's supply function for the New Haven market is given by:

$$P_S = 0.2 + 0.05Q$$

a) (4 points) What is the pre-tax equilibrium price and quantity?

Answer:

$$6.2 - 0.2Q = 0.2 + 0.05Q$$
$$6 = 0.25Q$$
$$Q = 24$$
$$P = \$1.4$$

b) (4 points) What is the (own-price) elasticity of demand at the equilibrium solved for in (a)? (Hint: Can answer in fractions)

Answer:

$$Q = 31 - 5P$$
$$\epsilon = -5 * \frac{1.4}{24} = -0.29$$

c) (5 points) The city has decided to impose a $0.20 per beverage tax. Solve for the new equilibrium price that consumers will pay.

$$P_S = 0.4 + 0.05Q = 6.2 - 0.2Q$$
$$5.8 = 0.25Q$$
$$Q = 23.2$$
$$P = 1.56$$

d) (2 points) What percent of this new tax is "paid" by consumers?

Answer: 16/20 = 80%

e) (2 points) If the goal of this tax was to decrease sweetened beverage by 10%, was it successful?

NO

Problem 4 (13 points)

Tesla offers two versions of its Model S sedan, the standard version and an "extended range" version. However, the two cars are physically identical; the standard version's software disables some of the onboard batteries to limit the range (this is a classic "damaged goods" strategy). So, the cost to manufacture either version is the same, approximately $50,000 per car.

After conducting some market research, they have identified two segments of car buyers: Luxury buyers, who have high reservation prices for a Tesla, and Eco buyers, who are more price sensitive and are considering competing electric vehicles as well. Reservation prices for the two types are as follows:

	Luxury	Eco
Standard Model S	70,000	60,000
Extended Range Model S	95,000	70,000

The firm estimates **there are 2 Eco buyers for every 1 Luxury buyer**.

a) (4 points) If Tesla were to only offer the Extended Range version, are they better off targeting the Luxury segment alone, or trying to attract both types of consumer?

Answer: attract both at a price of $70K. Profit is 20 per consumer == $60K per three consumers.

b) (5 points) If Tesla wants to offer both versions and have the types self-select into different cars, what are the best prices to charge for each version?

Answer: Charge 60K for the standard, and $85K for the Extended Range. Profit is 2*10+1*35=$55K per three consumers

c) (4 points) Is it worthwhile for Tesla to have the types self-select, or are they better off going with a single product (as in part a of this problem)?

Answer: better off in a.

Problem 5 (19 points)

A Yale undergrad has come to you for advice. They are graduating and looking at job options and have been offered two jobs, one that has a guaranteed salary of $85,000 per year, and the other that has a lower salary of $60,000 per year, but has a 75% chance of also having a $40,000 bonus.

This student is risk-averse and so is considering both opportunities. Their utility function is represented by $u(x) = \sqrt{x}$ where x is measured in thousands so that $\sqrt{\$100k} = 10$. They have no other savings or wealth.

a) (2 points) What is the expected value of the job with the bonus?

Answer: $90K

b) (5 points) Which job do you expect them to take?

Answer: $\sqrt{85} = 9.219$, while $0.25\sqrt{60} + 0.75\sqrt{100} = 9.436$
SO RISKIER JOB IS WHAT THEY WANT

c) (6 points) What is the minimum guaranteed salary the first job would have to offer to have this student prefer it over the risker job?

Answer: 89.04K.

$$\sqrt{x} = 0.25\sqrt{60} + 0.75\sqrt{100} = 9.436$$
$$9.436^2 = 89.04$$

d) (6 points) The firm offering the bonus has decided to offer a different contract for new analysts to accept. This new contract maintains the Expected Value of the existing contract, but put more weight on the bonus. Is this a good idea or bad idea when hiring potentially risk-averse employees, and why?

Answer: bad to go more high-powered if people are risk averse. Putting more of the salary in the bonus exposes the individual to more risk, and that this is costly to do if they are risk averse.

GAME THEORY

Problem 6: Extended location games [25 points]

Recall the location choice game discussed in the class: Tourists are evenly distributed on a beach that is one mile long. Two ice cream vendors simultaneously choose where to locate, each aiming to maximize their own market share. (Suppose they can locate exactly at the same spot if they want.) Tourists prefer to buy from the closer vendor. When they are indifferent between vendors, they randomly choose one. In the following, we consider a few variants of this location choice game.

Suppose that there are now **THREE** ice cream vendors.

(a) [5 points] Is it a Nash equilibrium that they all locate in the middle of the beach? Please explain your answer.

No, it is not a NE.

When they all locate in the middle, each vendor has a market share 1/3. But if a vendor individually deviates from the middles slightly, its market share can be close to 1/2.

(b) [5 points] Is it a Nash equilibrium that they locate at 1/6, 1/2, and 5/6, respectively (so that each has a market share 1/3)? Please explain your answer.

No, it is not a NE.

The vendor at 1/6 or 5/6 has a unilateral incentive to move toward the vendor in the middle. (The vendor at the middle actually has NO strict incentive to move.)

(c) [5 points] Suppose now we "bend" the beach so that it becomes a **CIRCLE** that has a circumference of one mile (i.e., the two endpoints 0 and 1 are now connected to each other). Is the location configuration in question (b) a Nash equilibrium? Please explain your answer.

Yes, it is now a NE.

Given the other two vendors' locations, a vendor's market share cannot exceed 1/3 no matter where it moves to.

Let us now return to the original location game with only **TWO** ice cream vendors and a **LINEAR** beach.

(d) [5 points] Suppose now each tourist never travels more than 1/4 mile to buy an ice cream. That is, if the distance to either vendor is longer than 1/4 mile, a tourist will not buy. What are the Nash equilibria now?

The only Nash equilibrium is that one vendor at 1/4 and the other at 3/4.

(e) [5 points] Suppose now each tourist never travels more than 1/5 mile to buy an ice cream. How will your answer change?

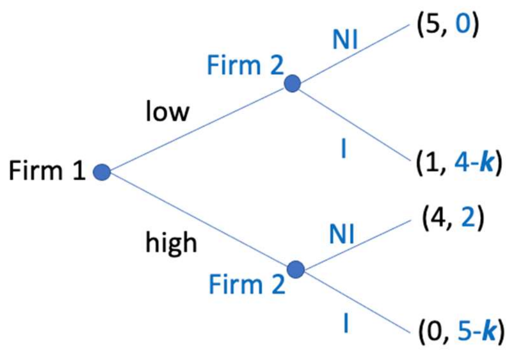
One Nash equilibrium is that one vendor at 1/5 and the other at 4/5. But now there are many other Nash equilibria. For example, one at 1/4 and the other at 3/4. In fact, any configuration is a NE where the two vendors are at least 2/5 mile away from each other and their distance to the closer endpoint is at least 1/5 mile.

Problem 7: Investment game [20 points]

Consider the following sequential-move game: firm 1 first decides whether to adopt an aggressive low-price policy or an accommodating high-price policy; after seeing firm 1's choice, firm 2 decides whether or not to invest in a new technology which reduces its production cost. The investment cost is $\boldsymbol{k}$.

If firm 1 sets a low price and firm 2 doesn't invest, then firm 1 will earn 5, and firm 2 will have no business (e.g., because its production cost is too high) and earn 0. If firm 1 sets a low price and firm 2 invests to reduce its production cost, then firm 1 earns 1 and firm 2 earns 4 (before considering its investment cost). If firm 1 sets a high price and firm 2 doesn't invest, firm 1 will earn 4, and firm 2 will have some business and earn 2. If firm 1 sets a high price and firm 2 invests, then firm 1 will have no business and earn 0, and firm 2 will earn 5 (before considering its investment cost).

(a) [5 points] Draw the game tree of this sequential-move game. [Hint: payoffs are net profit, so don't forget the investment cost.]



(b) [10 points] For what values of ***k***, will firm 1 choose to set a **high** price in the beginning?

k should be between 3 and 4.

Two conditions should be satisfied for firm 1 to set a high price:
First, when firm 1 sets a high price, firm 2 should respond by choosing not to invest. (Otherwise, firm 1 would get 0, which must be worse than choosing a low price.) This requires 5-***k***<2, or ***k***>3.
Second, when firm 1 sets a low price, firm 2 should respond by choosing to invest. (Otherwise, firm 1 would get 5, in which case it wouldn't choose a high price.) This requires 4-***k***>0, or ***k***<4.

(c) [5 points] Suppose the investment cost increases from ***k***=2.5 to ***k***=3.5. Will firm 2 suffer or benefit from that? Please explain your answer.

Firm 2 benefits from the increase of ***k***.

When ***k***=2.5, firm 1 will set a low price, and then firm 2 will invest, in which case firm 2 gets net profit 1.5.
When ***k***=3.5, firm 1 will set a high price, and then firm 2 will not invest, in which case firm 2 gets net profit 2. (A higher investment cost induces firm 1 to be less aggressive in the beginning.)